Information Delivery Specifications
IDS

BEAst Modelleringskrav - Tidiga skeden

Informationsleveranskrav baserat på BEAst Modelleringskrav – Tidiga skeden.

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Milestone: Tidiga skeden till och med Systemhandling.

Purpose: Framtagen för att ge olika parter som beställare, projektör och entreprenören en gemensam målbild för informationsleveranskraven på IFC modeller.

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Specifications

Projektets information

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCPROJECT**

that MEET the following requirements
- MUST HAVE attribute **Name** (Instructions: Project name)
- MUST HAVE attribute **Description** (Instructions: Project description)

Instructions
Project information and metadata required for all models

Samtliga objekt

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have

- value matching the pattern
  IFCBEAM|IFCBEARING|IFCBUILDINGELEMENTPROXY|IFCCHIMNEY|IFCCOLUMN|IFCCOURSE|IFCCOVERING|IFCCURTAINWALL|IFCDEEPFOUNDATION|IFCCAISSONFOUNDATION|IFCPILE|IFCDOOR|IFCEARTHWORKSELEMENT|IFCEARTHWORKSFILL|IFCREINFORCEDSOIL|IFCFOOTING|IFCKERB|IFCMEMBER|IFCMOORINGDEVICE|IFCNAVIGATIONELEMENT|IFCPAVEMENT|IFCPLATE|IFCRAIL|IFCRAILING|IFCRAMP|IFCRAMPFLIGHT|IFCROOF|IFCSHADINGDEVICE|IFCSLAB|IFCSTAIR|IFCSTAIRFLIGHT|IFCTRACKELEMENT|IFCWALL|IFCWALLSTANDARDCASE|IFCWINDOW|IFCBUILTELEMENT

that MEET the following requirements

- MUST HAVE property matching the pattern * of PSet **BIP**
- MUST HAVE property matching the pattern * of PSet **BaseQuantities**

Instructions

BIP-koder ska finnas på samtliga objekt

Våningstillhörighet

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have

- value matching the pattern
  IFCBEAM|IFCBEARING|IFCBUILDINGELEMENTPROXY|IFCCHIMNEY|IFCCOLUMN|IFCCOURSE|IFCCOVERING|IFCCURTAINWALL|IFCDEEPFOUNDATION|IFCCAISSONFOUNDATION|IFCPILE|IFCDOOR|IFCEARTHWORKSELEMENT|IFCEARTHWORKSFILL|IFCREINFORCEDSOIL|IFCFOOTING|IFCKERB|IFCMEMBER|IFCMOORINGDEVICE|IFCNAVIGATIONELEMENT|IFCPAVEMENT|IFCPLATE|IFCRAIL|IFCRAILING|IFCRAMP|IFCRAMPFLIGHT|IFCROOF|IFCSHADINGDEVICE|IFCSLAB|IFCSTAIR|IFCSTAIRFLIGHT|IFCTRACKELEMENT|IFCWALL|IFCWALLSTANDARDCASE|IFCWINDOW|IFCBUILTELEMENT

that MEET the following requirements

- MUST HAVE relationship **IFCRELAGGREGATES** with IFC class **IFCBUILDINGSTOREY**

Instructions

Alla objekt måste tillhöra en våning

Rum och utrymmen

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCSPACE**

that MEET the following requirements
- MUST HAVE property **Category** of PSet **BIP (IFCTEXT)** (Instructions: Lokalkategori)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCINTEGER)** (Instructions: Våningsplan)
- MUST HAVE property **NumberLant** of PSet **BIP (IFCINTEGER)** (Instructions: Lägenhetsnummer)
- MUST HAVE property **Name** of PSet **BIP (IFCTEXT)** (Instructions: Rumsnummer)
- MUST HAVE property **LongName** of PSet **BIP (IFCTEXT)** (Instructions: Rumsfunktion)
- MUST HAVE property **Description** of PSet **BIP (IFCTEXT)** (Instructions: Rumsnamn)
- MUST HAVE property **Reference** of PSet **BIP (IFCTEXT)** (Instructions: Typnummer (reference to RFP))
- MUST HAVE property **NetFloorArea** of PSet **BIP (IFCREAL)** (Instructions: Nettoarea)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 2.3

Byggdelar - Grundplatta/Bjälklag

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCSLAB**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning)
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material (if known))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **IsExternal** of PSet **BIP (IFCBOOLEAN)** (Instructions: Utvändig)
- MUST HAVE property **LoadBearing** of PSet **BIP (IFCBOOLEAN)** (Instructions: Bärande)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.2

Byggdelar - Golv

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCCOVERING**
- property **TypeID** of PSet **BIP (IFCTEXT) = UG01**

that MEET the following requirements
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)

Byggdelar - Yttervägg

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCWALL**
- property **IsExternal** of PSet **BIP (IFCBOOLEAN) = true**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (YVB01 for bearing, YV01 for non-bearing))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **LoadBearing** of PSet **BIP (IFCBOOLEAN)** (Instructions: Bärande)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.3

Byggdelar - Innervägg

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCWALL**
- property **IsExternal** of PSet **BIP (IFCBOOLEAN) = false**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (IVB01 for bearing, IV01 for non-bearing))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **LoadBearing** of PSet **BIP (IFCBOOLEAN)** (Instructions: Bärande)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.4

Byggdelar - Balkar / Pelare

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCBEAM**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (B01))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material (e.g., steel))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **LoadBearing** of PSet **BIP (IFCBOOLEAN) = true** (Instructions: Bärande (always true for beams))

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.5

Byggdelar - Balkong / Loftgång

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCSLAB**
- property **TypeID** of PSet **BIP (IFCTEXT) = BPL01**

that MEET the following requirements
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT) = 27.HC** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT) = Betong** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **IsExternal** of PSet **BIP (IFCBOOLEAN) = true** (Instructions: Utvändig (always true for balconies))

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.6

Byggdelar - Balkongräcke

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCRAILING**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (R101))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT) = 45.BB** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.6

Byggdelar - Trappor

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCSTAIR**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (TR01))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT) = 27.HB** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.7

Byggdelar - Undertak

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCCOVERING.CEILING**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (UT01))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT) = 43.E** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.8

Byggdelar - Tak

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCROOF**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (YT01))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT) = 41.A** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material)
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **IsExternal** of PSet **BIP (IFCBOOLEAN) = true** (Instructions: Utvändig (always true for roofs))

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.8

Byggdelar - Dörrar

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCDOOR**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (D01 for inner doors, YD01 for exterior doors))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material (e.g., wood))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **IsExternal** of PSet **BIP (IFCBOOLEAN)** (Instructions: Utvändig / Entrédörr (kan vara true eller false))

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.9

Byggdelar - Fönster / Glaspartier

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCWINDOW**

that MEET the following requirements
- MUST HAVE property **TypeID** of PSet **BIP (IFCTEXT)** (Instructions: Typbeteckning (F-01))
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material (e.g., aluminum))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)
- MUST HAVE property **IsExternal** of PSet **BIP (IFCBOOLEAN)** (Instructions: Utvändig (alltid true för fönster, false för invändiga glaspartier))

Instructions
Krav enligt BEAst Modelleringskrav kapitel 5.9

Byggdelar - Isolering

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCCOVERING.INSULATION**
- property **TypeID** of PSet **BIP (IFCTEXT) matching the pattern ISO.***

that MEET the following requirements
- MUST HAVE property **TypeDescription** of PSet **BIP (IFCTEXT)** (Instructions: Typbeskrivning)
- MUST HAVE property **BSABe** of PSet **BIP (IFCTEXT)** (Instructions: Klassifikationssystem (BSAB 96))
- MUST HAVE property **Material** of PSet **BIP (IFCTEXT)** (Instructions: Material (e.g., mineralull, stenull))
- MUST HAVE property **StatusConstruction** of PSet **BIP (IFCTEXT)** (Instructions: Byggnadsstatus)
- MUST HAVE property **StoreyName** of PSet **BIP (IFCTEXT)** (Instructions: Våningsplan)

Instructions
Requirements for insulation as specified in section 5.10

Våningsstruktur

IFC versions: **IFC2X3 IFC4**

The model MUST contain entities that have
- IFC class **IFCBUILDINGSTOREY**

that MEET the following requirements
- MUST HAVE attribute **Name matching the pattern \d3** (Instructions: Storey number (000, 010, 020, 025, etc.))
- MUST HAVE attribute **Description** (Instructions: Våningsbeskrivning.Plan 0 (Krypgrund), Plan 1 (Garage), Plan 2 (Entré), Plan 2 (Halvplan) osv.)
- MUST HAVE attribute **Elevation** (Instructions: Elevation from ground level)

Instructions
Requirements for building storeys as specified in section 2.2